Landsat Study of Land Surface Temperature Dynamics in Texas (2020 – 2024)

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Summary

Rapid urbanization in recent decades led to higher Land Surface Temperatures (LST) and reduced green spaces. Typically, urban heat islands have been estimated using ground-based observations, but nowadays remote sensing is widely being employed due to its broad spatial coverage and finer resolution. This study developed a remote sensing approach to monitor urban heating effects across the three most urbanized cities in Texas - Houston, San Antonio, and Dallas, by integrating satellite data with a fishnet-based zonal analysis from 2020 to 2024. Landsat 8 and 9 images were processed to derive LST, Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI), and a 75-km buffer was created for each city. A grid of 5 x 5 km was applied for zonal analysis to capture LST dynamics using a linear regression model to examine the relationships among LST, green vegetation indices, and built-up areas in the summers (June – August) and winters (December – February). Results show that Dallas experienced the highest summer LST, 47.5 °C in 2022, while Houston showed the lowest winter LST, 12 °C in 2024. LST rises in the summer; NDVI peaks in the summer and declines in the winter. The linear regression model reveals a positive trend between LST and NDBI in Dallas, whereas San Antonio shows no major trends during these periods. These findings underscore the impact of increased urbanization on green spaces and propose a robust remote sensing model to assess the overall heating effect.

Study Aims

This study aims to address the following research questions:

1. How can satellite remote sensing data (Landsat 8 and Landsat 9) combined with a cloud computing platform be used in effective urban heat mapping?
2. How has the Land Surface Temperature (LST) changed from 2020 to 2024 in the major cities of Texas?
3. What are the spatial and temporal trends of NDVI and NDBI in Texas over the past five years?

Methodological Framework

Three major urbanized cities in Texas (Dallas, Houston, and San Antonio) were chosen for this study, and a 75 km buffer around each city center was selected as the region of interest (ROI).

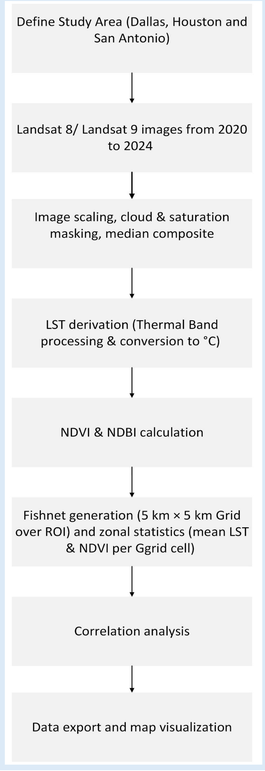
Landsat 8 (2020, 2021) and Landsat 9 (2022, 2023, 2024) Collection 2 Level-2 imagery for two periods, e.g., Summer (June to August) and Winter (December to February), between 2020 and 2024 was collected; less than 40% cloud cover was filtered and preprocessed by applying scale factors and cloud masking. A median composite was generated to minimize atmospheric noise.

LST data was derived, and NDVI was calculated using the red (SR_B4) and near-infrared (SR_B5) bands to identify vegetation health.

A 5 x 5 km fishnet grid was created over each region, and mean NDVI values were extracted for each grid using zonal statistics. Grids with NDVI > 0.5 were classified as dense vegetation zones.

NDBI was calculated using NIR and SWIR bands to establish a linear regression model with LST data.

The NDVI maps with fishnet grids were exported to ArcGIS Pro for final map preparation and appropriate visualization.



Results

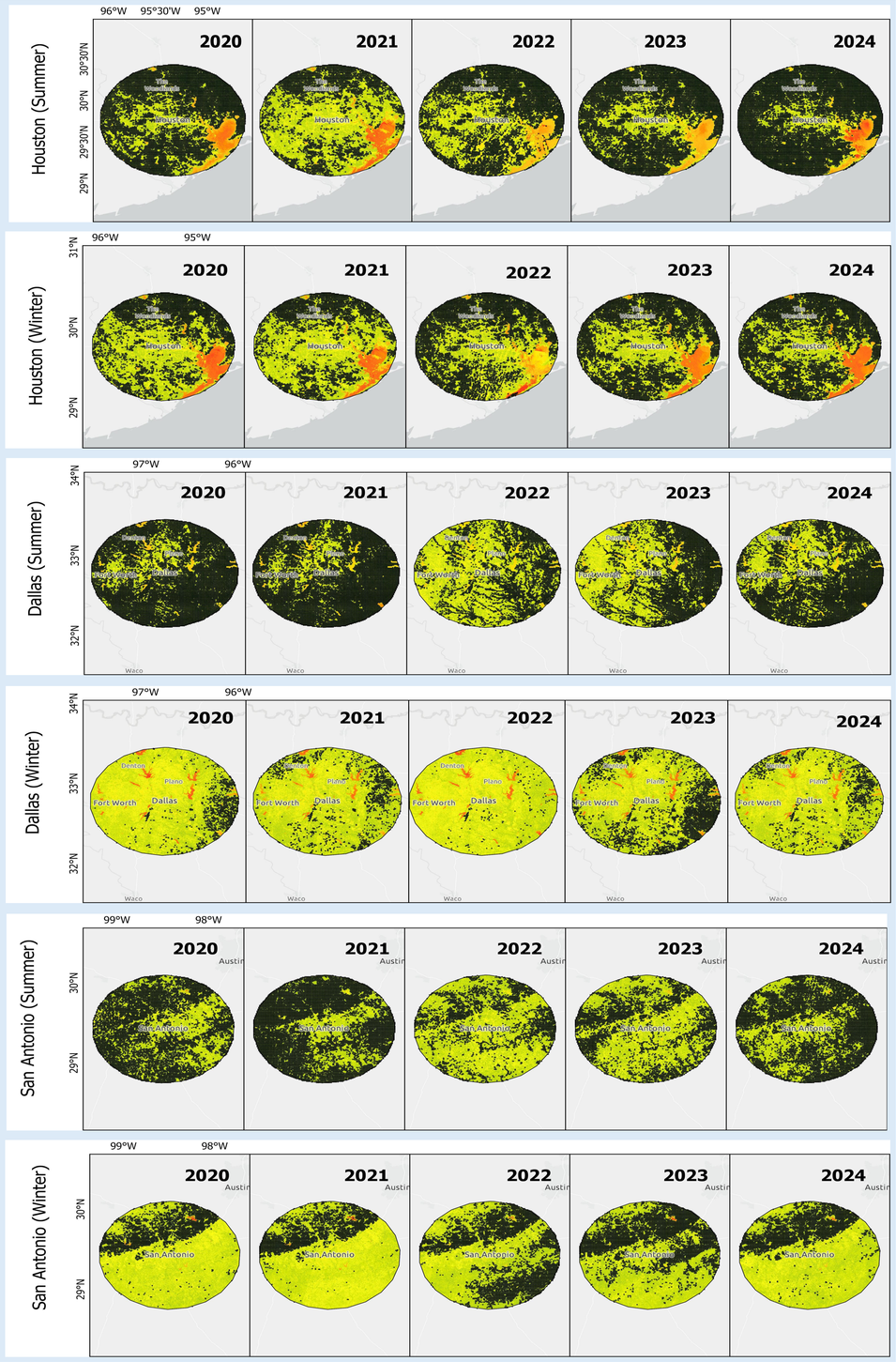


Fig. 1: Correlation between Land Surface Temperature (LST) and the Normalized Difference Vegetation Index (NDVI) in Texas using Landsat 8 and Landsat 9 data. A fishnet-based sampling method was applied to generate spatially distributed sample in relation to NDVI

Regression Outputs

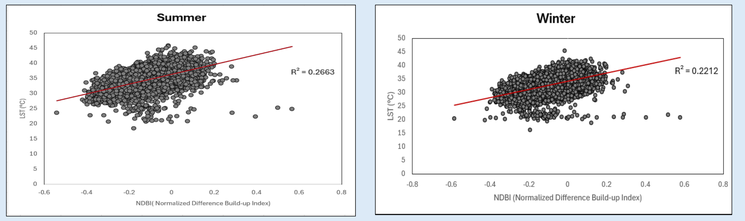


Fig. 2(a) : Correlation between LST and NDBI in Dallas from 2020 to 2024

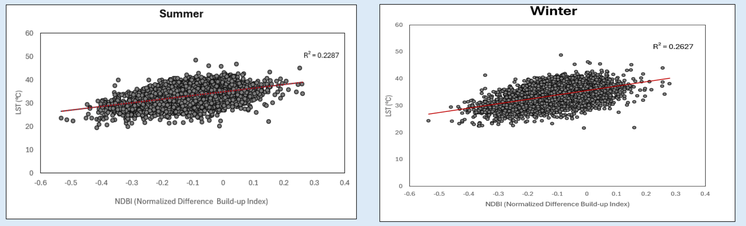


Fig. 2(b) : Correlation between LST and NDBI in Houston from 2020 to 2024

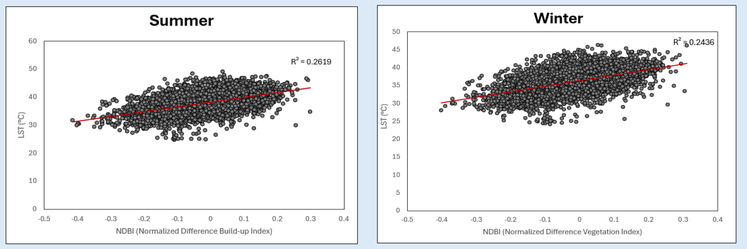


Fig. 2(c) : Correlation between LST and NDBI in San Antonio from 2020 to 2024

Fig.2: Linear regression outputs showing the relationship between LST and NDBI in major urbanized cities of Texas during summer and winter periods (2020–2024)

Conclusion

1. This study highlights the heterogenous trends of vegetation index in response to change in land surface temperature in the top three most urbanized cities in Texas (2020-2024).
2. Though demonstrating positive trends, the regression analysis between LST and NDBI reveals moderate relationship indicating other factors (e.g., vegetation cover, albedo, urban morphology, and anthropogenic heat) can play pivotal role in the increased temperature.
3. By incorporating Landsat data of a 30 m spatial resolution, this study proposes a robust approach for climate responding planning and urban landscape design.

Future Direction

1. Advanced Geospatial Artificial Intelligence (GeoAI) and Machine learning facilities will be integrated to establish connections among air pollution parameters, vegetation health and Urban heating Index (UHI) to better comprehend Urban Climate Dynamics over the past three years.
2. Future work also include developing predictive models for improved heat forecasting to support the development of greener space and heat resilient cities.

Acknowledgement

Many thanks to the Geoinformatics Research Group at the University of Arkansas. Thanks to the Department of Geosciences of the University of Arkansas for the travel grant. Thanks are also due to ESRI, USGS, NASA, Google Earth Engine for providing the necessary data to conduct this research.